

A Deep Learning Approach for Face Mask Recognition Using CNN Multi-model Architectures

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Abstract—During the COVID-19 pandemic, one of the most challenging hurdles was to control the corona virus from spreading and keep the number of individuals being affected at bay. As the number of infected cases increased, the healthcare resources became scarce making it a predicament for the authorities and healthcare providers. Due to the highly contagious and airborne nature of the virus, many nations have mandated the use of a face mask in public and crowded spaces as a preventive measure to minimize the virus from transmission. However, it is a strenuous task for the authorities to monitor huge masses to identify the individuals who do not wear the face mask and encourage the use of the same. This is where deep learning can play a vital role to ease the process of distinguishing people who wears a face mask from those who do not wear it. The aim of this project is to use deep learning techniques to recognize the face mask on individuals and report whether the person is wearing a face mask or not. For model implementation, a collaboration of Convolutional Neural Network (CNN) and ResNet architectures using transfer learning will be used. This fusion aims to increase the accuracy and robustness of face mask detection in a variety of real-world scenarios, addressing the critical need for effective public health initiatives. In vision techniques like this, image recognition and classification are the most crucial tasks, to train the model for achieving higher accuracy thus attaining the desired results.

Index Terms—CNN, ResNet, Image Classification, Deep Learning.

I. INTRODUCTION

The objective of this project is to distinguish the individuals who do not wear a face mask from those who do using a coalition of Convolutional Neural Network (CNN) and ResNet architectures along with transfer learning. This project not only focuses on classification of mask and non-mask users, but also thoroughly train the model to catalogue any other objects such as a coffee mug masking the face to be recognized and

classified as no mask. To achieve this accuracy, a dataset that contains approximately 7500 images with and without mask have been chosen as input. ResNet has a significantly stronger performance in transfer learning, and effectively handles the vanishing gradient problem. It is also well suited to handle complex tasks. As for the Convolutional Neural Network also known as CNN, it is a highly effective artificial neural network technique for image recognition. These two techniques are combined to create a transfer learning method that will be hugely efficient in detecting the face mask usage of an individual in the real-time environment. Due to its user-friendly and highly productive interface the entire architecture has been built in the Keras API on TensorFlow.

A. Motivation

The COVID-19 pandemic has wreaked havoc on the health, financial and psychological aspects of the individual and the society as a whole on a global scale. Bouncing back to normalcy was an arduous task for the government and healthcare authorities and the general public. A simple effort such as wearing a mask will be nearly 70% effective in prevention of the infection. The motivation of this project is to contribute in handling real-world problems using deep learning techniques, which in turn will facilitate the authorities to monitor and aid the massive population by recognizing individual non-mask wearers.

B. Research Question

How to distinguish real-world users who do not wear a face mask from the ones that do thus addressing the critical need for effective public health initiatives by building a transfer learning model using CNN and ResNet152V2 architectures?

II. RELATED WORK

To review the previous studies related to face mask recognition, assess the methodologies used, understand the limitations and choose the suitable deep learning methods for this project, a thorough research has been conducted on the existing studies.

Most of these studies have been conducted using Convolutional Neural Network (CNN) with architectures like Inception, ResNet, and MobileNet. CNN is one of best and effective artificial neural network algorithms for image recognition and classification. Some of the common and effective architectures in image recognition and classification are the YOLO5, ResNet, and Inception. In the following paragraphs, these studies have been reviewed in-depth.

To begin with, two comparative review papers have been examined to understand the techniques and requirements. This paper [1], a comprehensive review, provides an interesting insight about the list of applicable techniques, and their performance by comparing the results through meta-analysis. The authors not only compare the techniques, but also elicit the challenges, technical requirements, data requirements and so on to build a model with the optimal efficacy. Various attributes like image resolution requirements, quantity of data required to build a reliable and high performing model has also been discussed. It also suggests the quality of the outcomes to be achieved while tackling a real-world problem. The paper provides a substantial understanding to the research problem that is being handled in this project. Another comparative study [2], has simply compared few techniques based on their accuracy. Although it was good paper to start with, it was inadequate in terms of detailed analysis of all the factors to be considered which the first paper [1] has vividly executed.

To perceive the state of quality and quantity of the results that should be expected from a classification based deep learning techniques for the face mask recognition project, this paper [3] was quite helpful, as it covered a number of factors and different predictors is image classification. This paper introduced a dual-stage CNN model built with RetinaFace, Dlib, and MTCNN models. One limitation to this study was that it was not built on the new deep learning interfaces like TensorFlow that would have been more efficient.

Although this study based on the InceptionV3 model [4] gave an idea about transfer learning that our project aims to achieve, the output of the model was binary and it lacked thorough investigation of the other techniques. Despite being effective in terms of model building, the quality of the results were subpar. Insisting on the importance of the number of epochs to build a predictive model with high accuracy, this study [5] used YOLOv5 for face mask recognition. Apart from the knowledge on epochs in building a model, the authors failed to provide sufficient input for the image

classification model. With less than 900 images, this model was built to determine the whether the individual is with, without or incorrect mask on.

Interestingly, this research proposed a new model SSDMNv2, a DNN based detection algorithm using single shot multibox detector and MobileNetV2 [6]. It is to be noted that MobileNetV2 is more effective for mobile based applications. It is incorporated in OpenCV, Keras and Tensorflow. Eventhough careful considerations were carried out in the dataset selection which resulted in using a well rounded data as input and comparison on accuracy was clearly investigated, the research was focused on binary output contributing to insufficient outcomes while conventional methods produced qualifying results. A different study has its model built using Computer Vision [7] with Convolutional Neural Network (CNN) Algorithm, using three models such as ResNet50V2, VGG16, and MobileNetV2 trained by 20 epochs. This study shows that ResNet was slightly better in terms of performance and accuracy which was helpful this research. Though the model had a considerably sufficient input data, the model failed to differentiate the mask and other objects like a cloth or hand. The model failed to perform well in darker environments.

These two papers using YOLOv3 [8] and YOLOv7 [9] lacked detailing about the dataset which makes it difficult to understand the input parameters. Although the authors have claimed to improve the performance, failure of extensive research with other techniques and insufficient information about the input data lead to the uncut of the value the study especially in case of [9] where the results are not properly discussed.

Most of the CNN based studies like [10], [11], [12] and [13] have compared architectures like MTCNN and SVM like in [13], GoogleNet, VGGNet, ResNet in [10], RetinaFace MTCNN in [11] and VGGNet16 in [12]. A noteworthy effort in study [10] is that its usage of wide array of datasets that covered different shapes of faces to train the model for better prediction. Though, [11] claimed RetinaFace as the architecture with highest accuracy amongst its chosen architectures, there wasn't clear explanation about limitations as the model failed to poorly perform in crowded images. In the paper [12], the authors have failed to provide well informed comparison of methods and supporting visualizations in details which leads to inadequacy in evaluating the results and interpretations. This paper [13] also faces a similar problem like [12] where conclusions could have been more interpretable with quantifiable data. Despite of a few setbacks, these papers still support the use of CNN in image classification.

Using different methodologies on the same datasets helps in determining the advantages and disadvantages of each methodologies with respect to accuracy, performance,

optimization and hardware requirements. This where these papers come into picture. The first paper is a modular approach [14] that compares RetinaFace, MobileNet, PyramidKey and MTCNN while the second [15] is a multi-stage image processing using YOLOv3, Faster R-CNN and customized CNN. Both the studies work with 3 large data which provides remarkable inputs to the models. While Faster CNN is concluded as the better model in the multi-stage image processing model [15], the modular approach [14] claims RetinaFace as a better choice. However, it should be understood that there are no coinciding models, hence the authors' perspective is limited on this front.

As ResNet152v2 handles the vanishing gradient problem when it comes to image classification using residual neural networks, and MobileNetV2 has been widely used in classification problems to provide remarkable accuracy, few other papers related to image classification using ResNet152v2 and MobileNetV2 architecture have been researched to obtain crucial knowledge on the subject.

In this study of Automatic Classification of Alzheimer's [16] brain MRI is used to classify the presence of Alzheimer's. ResNet152v2 being the pivotal architecture for this study, the authors have claimed that the architecture was highly effective in capturing even the smallest of the anomalies. It is noteworthy that the authors have concluded that ResNet152v2 performed exceptionally well in filtering the brain patterns. The authors conclude by saying that there was a impressive precision in the predictions.

Another transfer learning approach paper proved quite beneficial for this project was the study to detect plan leaf disease [17] using EfficientNetV2L, MobileNetV2, and ResNet152V2. Although the authors conclude that EfficientNetV2L is the best amongst the architecture, the difference is as mere as 0.5% and 1% for MobileNetV2 and ResNet152V2 respectively. Also, MobileNetV2 is aimed mainly for mobile based applications. In this study for Diagnosis of Diabetic Retinopathy [18] with two datasets to detect five different degrees of the diagnosis, four architecture such as ResNet50, ResNet152V2, VGG16 and VGG19 were compared and ResNet152V2 was the second best method of all. However, the technique required only half to one-fourth of the epochs to attain better results of what the other architectures required. Additionally, in this research to detect wild animals using DCNN and ResNet152V2 [19], the results provided 97% accuracy in classifying the wildlife.

Finally, a closely related paper [20] to this project in face mask detection using CNN and DNN with architectures like MobileNetV2, Xception and ResNet152v2 confirmed that ResNet152V2 achieved 99.80% accuracy and MobileNetV2 achieved 99.51% accuracy. The model [20] is also built using Keras in the TensorFlow platform which is quite similar to this project.

Although all these studies have either adapted a collective of methods or individual architectures, they emphasize on the importance of deep learning in solving real life problems. Most of these conundrums can be handled effectively using deep learning and machine learning techniques.

For this project, through the knowledge gathered from the previous researches, it has been inferred that Convolutional Neural Networks (CNN) with ResNet152v2 and MobileNetV2 architectures are highly effective for image classification and recognition. From the paper [1] which is a comparative study and the papers [16] and [20] that strongly support the usage of the aforesaid architectures, it could be gathered that, Convolutional Neural Network (CNN) with ResNet152v2 and MobileNetV2 architectures are some of the optimal choices to address the problem background and answer the research question.

These studies not only served as a knowledge bank for deriving information about the deep learning techniques, but also to gain a deeper insight on the pre-requisites like data pre-processing, data requirements, hardware requirements and so on. For example, papers like [18] and [1] explained the methodologies and input requirements to achieve high accuracy and performance in length. Though few papers appear to have limited resources, they were still able to provide detailed explanation of the chosen methodologies.

III. DATA MINING METHODOLOGY

In this analysis Knowledge Discovery in Database (KDD) framework has been employed as a foundation to extract the useful knowledge and insights about the face mask detection. KDD offers a systematic and structured approach consists of various stages as shown in Fig. 1, such as data collection, pre-processing, transformation, modelling, evaluation and interpretation. The author carefully selected the dataset and pre-processed the dataset to ensure its quality and relevance to research objectives. This has been depicted in Fig. 1.

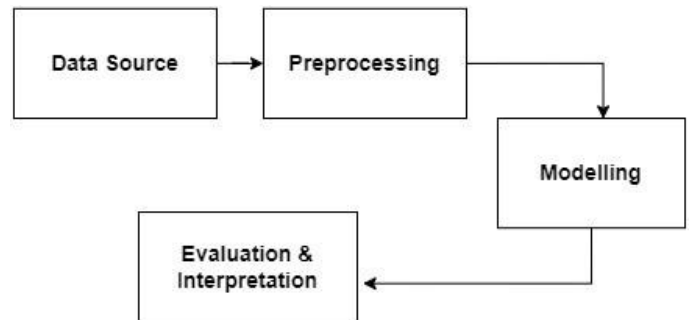


Fig. 1. KDD Framework.

Through careful exploration and analysis deep learning techniques like Convolution Neural Networks (CNN) with ResNet152V2, and MobileNetV2 architecture has been used. In order to achieve optimal performance in face mask detection

based on a wide range of real world circumstances, the modelling phase has been concentrates on training and optimizing the models. Rigorous evaluation metrics and methodologies are used to asses the effectiveness and robustness of the approach. Finally, the interpretation stage involves deriving actionable insights and practical recommendations from data to advance the knowledge on face mask detection system and its results for public health initiatives during COVID-19 pandemic.

A. Data Selection and Dataset Description

Input dataset is extremely important to train a deep learning model, especially in image classification. As data selection is an integral part of building deep learning models, the process was carried out with meticulous due diligence. This dataset from Kaggle [21], has ample resources for both with and without mask images. Comprising of 7553 images in total, out of which 3725 images showing individuals wearing masks and 3828 individuals without a face mask, this dataset provides a diverse dataset inclusive of a varied ethnicity, gender, age, additional accessories, orientation of the individual in regards to the camera angle, monochromaticity, different colours and types of masks, and other objects like coffee mugs masking the face. All these are RGB images with three colour channels. Few examples of the input data have been depicted in Fig. 2 and Fig. 3 for with and without mask categories respectively. This is an open dataset hence there are no privacy or ethical concerns in using the dataset for this research.



Fig. 2. Sample Input Data for With Mask.

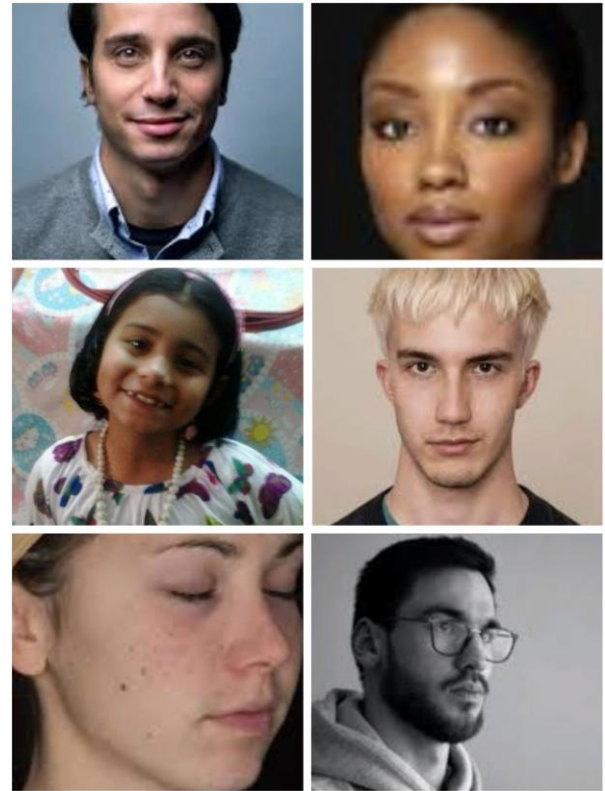


Fig. 3. Sample Input Data for Without Mask.

B. Data Pre-processing

Data pre-processing is a crucial step in training a model using the image processing technique, aiming to ensure quality, consistency and suitability. This process involves several steps to clean and prepare the data for analysis.

Initially a empty list has been created as dataset to store the pre-processed images. After that, iterating through “mask_files” and “not_mask_files” which is set of image files, it is assumed that these contains the pictures of people who are wearing masking and not wearing masks respectively.

Steps carried out in data pre-processing:

- The image files are opened using Image module from the Python Imaging Library (PLI).
- The images has been resized to a fixed size of 128x128 pixels, which is a common pre-processing step to ensure uniformity in dataset.
- The images are then converted to RGB colour space if it is not already in that format.
- Then the images are transformed into a NumPy array, a common format for image data in machine learning applications.
- Finally, it appends the pre-processed images to list called “dataset”.

The above steps will be executed for the list of images which contains both people wearing and not wearing masks. Once all the images have been pre-processed it is split into train and test data, with 80% of the data allocated for training the model and 20% of the data is used to test the model. By splitting the dataset in this manner, the models can be effectively trained and evaluate the machine learning models on separate subsets of data and it facilitates the assessment of its performance.

C. Model Implementation

In the model implementation phase, data pre-processing efforts are converted into actionable insights by training the machine learning algorithms on the prepared dataset. This stage involves the implementation of the selected models such as Convolution Neural Networks (CNN) and ResNet architecture for building Transfer Learning model to effectively classify images as either depicting individuals wearing face masks or not. Using this model, the aim is to develop robust and accurate system which capable of detecting the face mask with high precision and recall. The model implementation phase addresses the critical need for effective public health initiatives to overcome the spread of COVID-19 virus.

1) Convolutional Neural Networks (CNN): Convolutional Neural Networks (CNN) is a popular class of deep learning model which are widely used for image recognition and classification tasks. CNN models are composed of several layers and modelled by inspiring human brain's visual cortex. These sequential layer includes Convolutional, pooling, and fully connected layers. These networks are proficient in automatically extracting hierarchical features from the input images, which allows to find accurately whether the individuals are wearing mask or any other objects covering the face is not consider as wearing mask.

The first layer present in this model is a Convolution layer, which is a fundamental components of CNN responsible for extracting features from input images through convolutions with learnable filters known as kernels. In this model, two convolution layers are added followed by a Rectified Linear Unit (ReLU) activation function. The model gains non-linearity from the ReLU activation function which helps to identify the complex patterns in the data. The input images are subjected to 32 filters of size 3 x 3 in the first convolution layer and 64 filters of the same size in the second layer. By using these filters the model has the ability to identify various patterns and trends present in the input image.

After each convolution layer a max pooling layer has been added, which plays a crucial role in minimizing the spatial dimension of the feature maps generated by the convolution layers. Max pool layers down sample by opting the maximum value within each features of the map region, it also lowers the computational complexity and avoids over-fitting. In this model, 2x2 pooling windows have

been used to down sample.

Following the convolution and pooling layer, the model is enhanced by adding a flattening layer which transforms the multidimensional output of the previous layer into a one dimensional array. It serves as an input to the next coming fully connected dense layers. This process is necessary for transition from the convolution layer to dense layer.

In this model, two fully connected dense layers are used these layers are traditional neural network layers where each neuron is connected to all the neurons in the previous and subsequent layers. These layers capture complex patterns from the feature maps and enabling the model to learn a high-level representation of data. In the model, the first dense layer consists of 128 neurons and second layer consist of 64 neurons with the same activation function ReLU.

To prevent the over-fitting, Dropout Regularization is applied after each dense layer at the rate of 0.5. Dropout randomly deactivates a portion of neurons during training, forcing the model to learn unnecessary representation and minimizes the risk of over-fitting the training data.

The output layer of the model consists of two neurons with Sigmoid activation function. The Sigmoid activation distinguishes the output values between 0 and 1, as either the individual is wearing mask or not. Therefore, ReLU is used in the intermediate dense layer to produce non-linearity and capture complex patterns in data and Sigmoid function is used in output layer to produce probability score for binary classification. The combination of ReLU and Sigmoid function is commonly used in neural network architecture which helps in effective learning and the model's capacity for prediction.

The model is compiled using three key parameters such as optimizer, loss and metrics. The optimizer has employed Adam which is an adaptive learning rate optimization algorithm, it is used to update the model's weights during training to minimize the loss. The loss component employed sparse categorical cross entropy loss function, which is suitable for integer encoded target labels. To gather knowledge on performance with respect to the number of epochs, accuracy has been incorporated as a metric to evaluate the model.

2) Residual Network (ResNet152v2): Residual Network (ResNet152v2) is introduced to overcome the deep learning issue of vanishing or exploding gradient while adding more layers to the CNN architecture for any classification problem. Here applied a method to skip connections in the network. It bypasses certain levels in between to link-layer activation to other subsequent layers. The leftover blocks are stacked over to create the ResNets. This in-turn increase the accuracy and performance of the model.

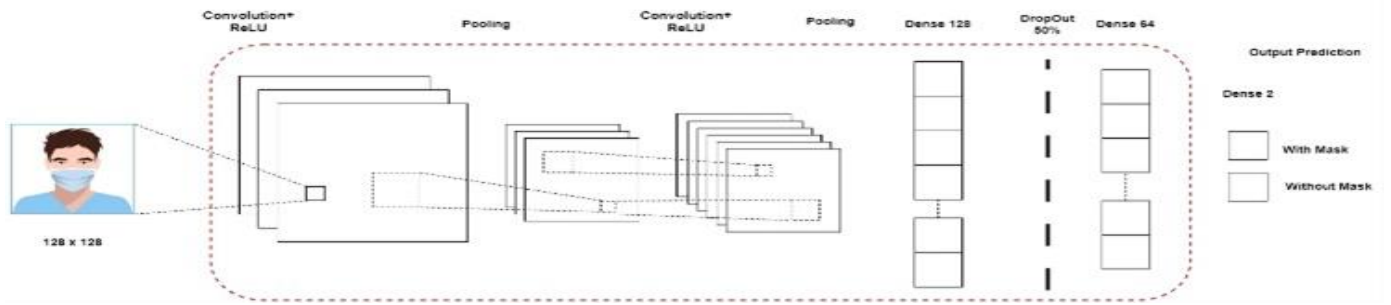


Fig. 4. Convolutional Neural Networks (CNN).



Fig. 5. Residual Network (ResNet152v2) Architecture.

In this problem of image classification of wearing a mask or not, the base model is pre-trained with the ResNet152V2. Since this model is a much deeper network compared to the traditional Convolutional Neural Network, having 152 layers enable the model to learn complex features and patterns from the dataset. Also, skipping the connections aids in addressing the issue of vanishing gradient problem during the training of the model. In this case, the model leverages the use of pre-trained dataset of 'ImageNet'. Use of these pre-trained weights enable to capture features that is already learned from a diverse set of images. This can be highly advantageous during image classification problem with limited data source, where fine-tuning the model aids in increasing the performance than using traditional models that need to learn from scratch.

The ability to capture detailed features and patterns from the dataset helps to have the knowledge to be used on unseen data. In this problem of creating a deep learning model for mask detection in real world scenarios, faces may appear in different orientations, different lighting conditions, or with different colour of masks. ResNet152 offers several advantages to understand the dataset with the pre-trained weights that making better prediction in classifying a picture whether a mask is present or not.

In this proposed model, the pre-trained model with the

ImageNet weights, creates an output and the ResNet152V2 acts as a feature extraction tool to understand and evaluate the initial features from the pre-processed image in the dataset. After the features are extracted, the output is flattened to convert the base-model to a 1-Dimensional tensor. Following this, a batch normalization layer is added to the model to normalize the activation of the previous layer. This helps in addressing the internal covariate shift during training.

During the normalization of the activation, this accelerates the training process by allowing higher learning rates. This helps to make the model more powerful and robust to parameter initialization and act as a form of regularization. This also fixes the issue of over-fitting. This normalization helps in stabilizing and also the training of deep neural networks by normalizing the inputs to each layer.

Following this normalization, a dense layer is added to the model with 128 units and ReLU activation function signifying the model has 128 neurons and each neuron will apply ReLU activation function to each input of this neuron. In this model ReLU activation function is used to help the neural network learn more complex patterns and representation from the dataset. This introduces non-linearity to the neural network by taking the input directly.

The next comes the dropout layer to regularize the model and prevent over-fitting of the dataset to the model. This is carried out as a process of regularization that is achieved by randomly setting a fraction of units to zero to each update of the model parameters. Eventually, during the dropout layer 20% of the previously trained neurons are dropped from the model. This is required as a process in building the Convolutional Neural Network to prevent the network relying too much of specific details in certain neurons and in turn enhances other neuron to learn better features that are robust. It also helps in over-fitting and generalization of the model.

Finally comes another dense layer with 2 units having a Sigmoid activation function to find the image by performing a binary classification. The class generated is 2 and these units represent the probabilities of the input data belong to each class. Sigmoid function is used limiting the range of

output values between 0 and 1 which can be interpreted for the values from the input belonging to the positive class for each unit.

The designed model is compiled and is optimized with a learning rate of 0.001. the proposed model uses a variant of the Adam optimizer, Adamax. The learning rate decides the step size to be calculated for the optimization, also on how quick the model learns and the granularity of the adjustments required to tune the model.

Since this is a problem of classification task, `sparse_categorical_crossentropy` is used as the loss function. The dataset is labelled with 0 and 1s in the form of integers. Accuracy is used as the metrics to be used to classify the samples. This helps to measure the performance of the classification task that is performed. Thus the overall Convolutional Neural Network model following a transfer learning from the ResNet152V2 as a feature extractor followed by a fully connected layer is built for the classification of an image having a face mask.

The training of the built model is done by keeping the epochs count as 10, meaning the number of iterations done over the entire training dataset. Also, a validation split of 10% is considered to evaluate the data validation after each epoch run. This is performed to analyse the model performance and the possible situation of over-fitting.

3) *MobileNetV2*: Another model using MobileNetV2 has been built to compare and validate accuracy of the results. The input data has been split into train and test data with 80:20 ratio respectively. To process the data for image classification the input is scaled to shape images with 128*128 dimensions. To implement this model, the libraries used in this project are Sequential, Flatten, Dense, Dropout, BatchNormalization and Adamax. They are explained in detail in the upcoming sections.

To create the model using linear stack of layers, Sequential framework has been used from Keras, TensorFlow. In this method, each layer has only one input and output tensor. Multiple layers can be added using `add()` function. The type of layers used in this model are Flatten, Dense, Dropout, BatchNormalization. They are described as follows:

- Using Flatten, the multi-dimensional data is converted into one-dimensional array.
- The Dense layer is where the neurons execute the weighted sum, adds bias and uses activation function to create the output. Hence this is also known as the fully connected layer.
- To randomly drop inputs, Dropout layer is used. For example, setting this to 0.3 means 30% of the input is drop. The Dropout layer usually follows the Dense layer.

- Batch normalization is where the scaling and shifting happens in order to increase the performance of the model.

For optimization and model training, the Adamax extension from Adam optimizer has been used. It is highly effective on deep neural networks. Using `compile()` function, optimizer can be incorporated. It is adaptive and quick. Two models have been built for this architecture. One base model and one improved model. There are three input variables. They are,

- `'image_size'`: The dimensions for the input image is defined as 128*128 for using this variable.
- `'channels'`: This variable specifies the number of colour channels, which is 3 in this case.
- `'image_shape'`: This parameter defines the shape of the input images by combining the first two variables.

The base model has been built using MobileNetV2. This is where the MobileNetV2 architecture is initiated with pre-trained weights from the ImageNet dataset. Once this model has been created, it can be further extended into additional layers.

Once the base model is created, using Sequential framework, it is extended into building linear layers to the model. Here the Flatten(), BatchNormalization(), Dense() and Dropout() layers are defined. In the Dense layer, additional fully connected layer with 128 input units has been added. The activation function ReLU has been used for the hidden layers as it handles the vanishing gradient problem well. For the output layer, Sigmoid is used as the activation function, as it is well suited for binary classification.

To further fine tune the model, the trainable weight is set to false to prevent the pre-trained weights from getting altered by the gradients. While dealing with large datasets, it is essential to maintain the integrity of the model without having to recreate the model. This can be used by using MobileNetV2 model with pre-trained weights and freezing its layers.

The model is then optimized using Adamax optimizer. This has been Incorporated using the `compile()` function. The learning rate is 0.001. As for the loss function `'sparse_categorical_crossentropy'` has been used as it works well for multi-class classification problems with integer labels. To get the accuracy of the model, the metrics is set to `'accuracy.'` The summary of the model is shown in Fig. 5.

Layer (type)	Output Shape	Param #
mobilenetv2_1.00_128 (Functional)	?	2,257,984
flatten_2 (Flatten)	?	0 (unbuilt)
batch_normalization (BatchNormalization)	?	0 (unbuilt)
dense_4 (Dense)	?	0 (unbuilt)
dropout (Dropout)	?	0 (unbuilt)
dense_5 (Dense)	?	0 (unbuilt)

Total params: 2,257,984 (8.61 MB)

Trainable params: 2,223,872 (8.48 MB)

Non-trainable params: 34,112 (133.25 KB)

Fig. 6. Model Summary - MobileNetV2.

To further optimize the model, Adam optimizer has been used and for the loss function 'categorical_crossentropy' is used. It works better where the labels are one-hot encoded.

Next step is to train and fit the model. The epoch is set to 5 to iterate the model training five times. 10% of the train data is set for validation by setting it to 0.1. This is where the model, trains itself to classify the input data.

In the next step, the accuracy of the model has been validated. The model has a remarkable accuracy of 99%. This has been illustrated in Fig. 6.

48/48 ————— 38s 799ms/step - accuracy: 0.9900 - loss: 0.1188
Test Accuracy = 0.9900727868080139

Fig. 7. Model Accuracy - MobileNetV2.

For further validation the outputs have been plotted which will be explained in the Results and Interpretation section.

IV. EVALUATION AND RESULTS

The evaluation and results section demonstrates how the chosen methodology Convolution Neural Networks (CNN), ResNet152v2 and MobileNetV2 has been applied to address the research question, it assesses the model's efficiency and discuss about the potential effects of the findings. This section explains about the evaluation methodology, performance metrics, parameterization choices, and outcomes.

Choice of Model Parameters in Proposed Models:

The model that needs to be trained with the dataset is built with a basic model and few other parameters are used to extract maximum feature to train the model. In all 3 models built, the image in the dataset is resized and scaled to the required size. After the pre-processing of the image dataset is completed, the train split is fed into the base model built. However, for the model built using Resnet152v2 and MobilenetV2, feature extraction is done using the above mentioned extractor followed by building a Convolutional

neural network model which include the batch normalisation, a dense layer with 128 units which has a ReLU activation function. Using the function helps to introduce non-linearity to the network and also sparse activation, meaning only a small subset of neurons is activated for any given image input.

Followed by this is a dropout layer of 0.2 used throughout the model. This is picked as an ideal choice as 20% of the neurons are dropped. This rate is chosen to prevent over-fitting of the model. Keeping the rates lower than this leads to train the model with maximum exposure and leading to bias and over-fitting.

In compiling the model, the adamax optimizer is used for training the neural network since it tends to converge faster compared to adam optimizer. It is also chosen due to its efficiency with memory, simplicity and robustness.

Model trained with other learning rate, type of optimizer and dropout rate tend to give lower accuracy for the models predicted using the trained model. The Resnet and ImageNet models for the above parameters are trained with this setting gives accuracy higher compared to other models with different parameters.

A. Convolution Neural Networks (CNN)

In evaluating Convolution Neural Networks (CNN) a range of performance metrics has been used to assess the built model's effectiveness. Common metrics for classification tasks are Training & Validation Performance plot, confusion metrics and classification metrics include accuracy, precision, recall, and F1-score. These metrics provide insights about the different features of the model in classifying whether the individual is wearing mask or not and its overall predictive power.

Training and Validation Performance plot visualize the model's performance over the epochs during training. It helps to monitor the training and test accuracy/ loss trends to detect the over-fitting or under-fitting. Accuracy for training starts at 76.54% in 1st epoch and steadily increases to 97.23% by 10th epoch. Accuracy for validation starts at 86.05% in 1st epoch and shows fluctuations before it settles at 92.83% in 10th epoch. Both the training and validation accuracy shows an increasing trend over the epochs, indicates the model is learning and improving its performance.

Similarly, the loss for training starts at a high value of 55.92% and decreases consistently to 7.06% by the 10th epoch. Loss for validation starts 31.23% in the 1st epoch and decreases steadily to 28.12% in the 10th epoch. Both the training and validation losses exhibit downward trends over epoch, states the model is effectively minimizing its error.

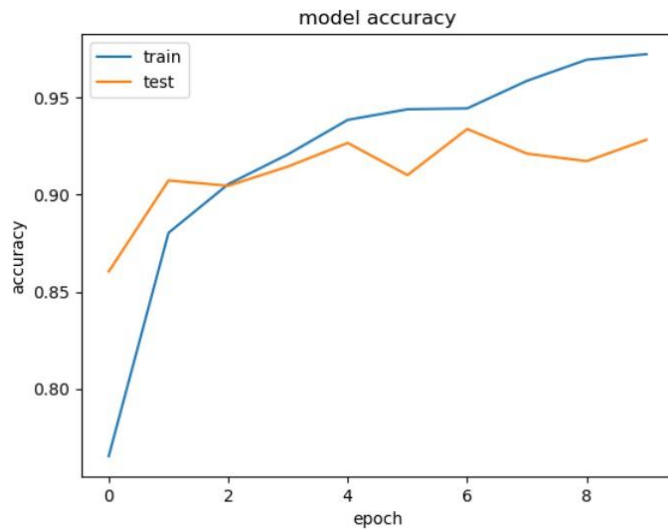


Fig. 8. Accuracy Plot - CNN.

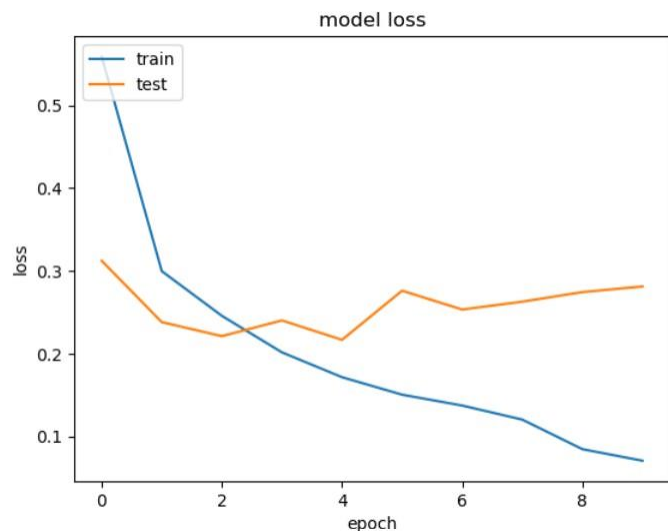


Fig. 9. Loss Plot - CNN.

Overall, there are no significant signs of over-fitting or under-fitting seen in the model according to the training and validation performance plot. Which indicates that the model is successfully learning from the training data and predicting well for the test data.

Confusion Matrix is to evaluate the model by summarizing the number of correct and incorrect predictions for each class, it represents the model's ability to classify data correctly.

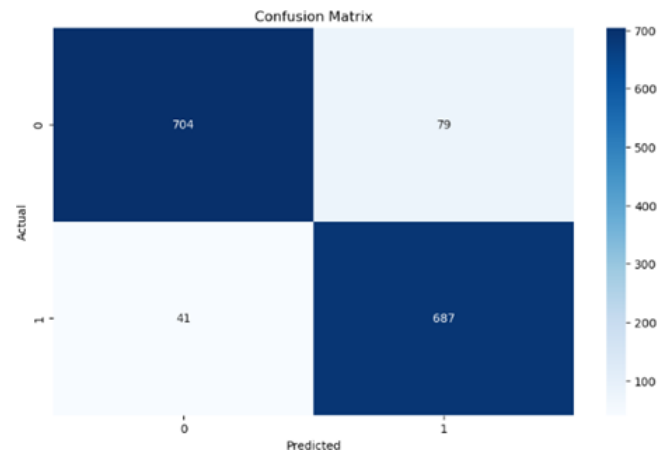


Fig. 10. Confusion Matrix - CNN.

Fig. 10 shows the confusion matrix with class 0 (Not Wearing Mask), there are 704 true negatives, indicates correctly the individuals are not wearing mask. Conversely, there are 79 false positives instances, where incorrectly classified as wearing a mask when they are not actually wearing. On the other hand, class 1 (Wearing Mask), 687 true positives demonstrates it correctly identified that the individuals are wearing mask. Also, 41 false negatives instances identified incorrectly where the individuals are not wearing mask when they were wearing. This breakdown shows the model's effectiveness in distinguishing between the classes.

Fig. 11 shows the classification report for the Convolution Network Models (CNN), it demonstrates the strong performance metrics like below on the testing data,

	precision	recall	f1-score	support
0	0.94	0.90	0.92	783
1	0.90	0.94	0.92	728
accuracy			0.92	1511
macro avg	0.92	0.92	0.92	1511
weighted avg	0.92	0.92	0.92	1511

Fig. 11. Classification Report - CNN.

- **Precision:** It measures accuracy of the positive predictions made by the model. For class 0, where the individuals are not wearing mask the value is 0.94. Which means 94% of the instances predicted as not wearing mask , are not actually wearing mask. On the other hand, for class 1 where the individuals are wearing mask the value is 0.90, which means that 90% of the instances predicted as wearing mask are actually wearing mask.
- **Recall:** It measures the model's ability to capture all positive instances in the data provided. For class 0, where the individuals are not wearing mask the value is 0.90. Which means 90% of the instances predicted as not wearing mask , are not actually wearing mask. On the

other hand, for class 1 where the individuals are wearing mask the value is 0.94, which means that 94% of the instances predicted as wearing mask are actually wearing mask.

- **F1-Score:** It is the mean of both precision and recall by providing single value that balance the both metrics. This model has provided a value of 0.92, which indicates a 92% balanced performance between precision and recall.
- **Accuracy:** It is a key metric used to evaluate the overall performance of a classification model. This model provides a accuracy of 0.92, which means the model as 92% of the ability to classify whether the individual is wearing mask or not.

To summarize, despite being 92% accurate, the CNN model can be trained further to extract maximum features from the images by employing advanced architectures like ResNet152V2 and MobileNetV2 which can be used to develop a transfer learning model for classification.

B. Residual Network (ResNet152V2)

The evaluation methodology for Residual Network (ResNet152V2) method involves training the model on a dataset of labelled images and validation the same on the separate test dataset. Then the metrics are evaluated using the following techniques.

The training accuracy steadily increases from 89.75% to 99.45% over the epoch, indicates the model is learning and improving its performance over the time. Similarly, the training loss decrease 32.75% to 1.85% showing the models predictions are more accurate.

The validation accuracy also follows the same pattern, increasing from 97.02% to 98.35%, indicates the model's performance on unseen data is improving with training. Similarly, the validation loss decreases from 21.28% to 9.1% indicates the model's predictions are more accurate.

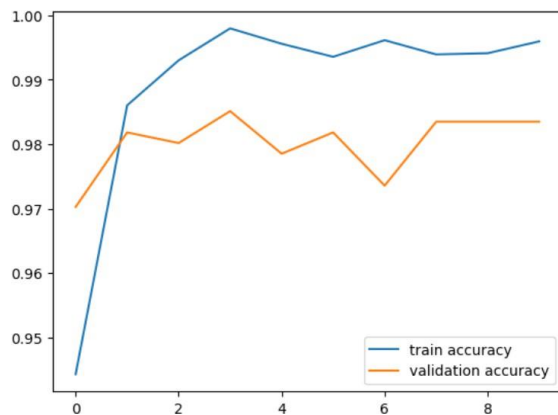


Fig. 12. Accuracy Plot - ResNet152V2.

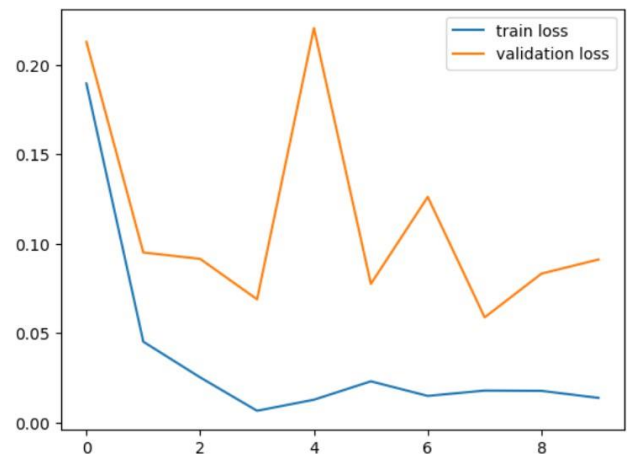


Fig. 13. Loss Plot - ResNet152V2.

The training and validation curves are almost similar to each other, suggesting that the model is not over-fitting or under-fitting the training data. The consistency between two curves indicates that the model is successfully learning from the training data and predicting well for the test data.

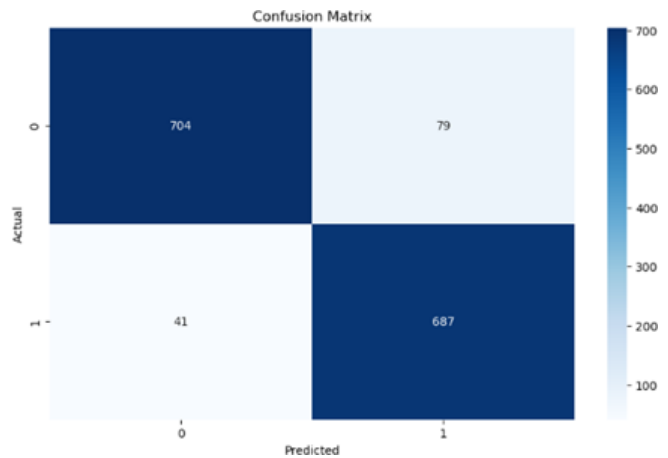


Fig. 14. Confusion Matrix - ResNet152V2.

Fig. 19 shows the confusion matrix with class 0 (Not Wearing Mask), there are 741 true negatives, indicates correctly the individuals are not wearing mask. Conversely, there are 9 false positives instances, where incorrectly classified as wearing a mask when they are not actually wearing. On the other hand, class 1 (Wearing Mask), 752 true positives demonstrate it correctly identified that the individuals are wearing mask. Also, 9 false negatives instances identified incorrectly where the individuals are not wearing mask when they were wearing. This breakdown shows the model's effectiveness in distinguishing between the classes.

Fig. 20 shows the classification report for the Residual Network (ResNet152v2), it demonstrates the strong performance

metrics like below on the testing data,

	precision	recall	f1-score	support
0	0.94	0.90	0.92	783
1	0.90	0.94	0.92	728
accuracy			0.92	1511
macro avg	0.92	0.92	0.92	1511
weighted avg	0.92	0.92	0.92	1511

Fig. 15. Classification Report - ResNet152V2.

The model achieved a precision of 94% for instance to classify as not wearing mask and 90% as wearing mask, which indicates majority of the prediction done by the model are accurate. The recall value of 90% for class 0 and 0.94 for class 1 demonstrate the models to correctly classify the categories. The F1-score yielded consistent score of 92% represents a balanced performance between precision and recall. Overall, accuracy of the model is 92%, which represents the model ability to find whether the individual is wearing mask or not.



Fig. 16. Result - ResNet152V2.

Fig.21 shows the prediction of individuals wearing mask or not, however this model's prediction was less accurate where an object is brought close to the face. It considers the objects as mask and predicting wrongly. To overcome this classification problem, another architecture called MobileNetV2 has been implemented.

1) *MobileNetV2*: For the MobileNetV2 model, the initial features are extracted by training with the image dataset. The pre-processed image dataset is used for the feature extraction and further sent to the further dense layers and dropout layer as designed for the model. For the given hyperparameters, the model is trained with 5 epochs. This is selected after training with other validation runs and getting the better accuracy scores for the selected epoch runs. The entire training dataset is iterated over 5 times and each epoch consists of one forward and one backward pass for each sample of the dataset.

The performance of the trained model is evaluated against the test dataset to understand how well the prediction is using training validation and accuracy plots, confusion matrix and classification report.

From the plots generated for Training and Validation accuracy and loss, the model has learned progressively over the epochs, where the losses decreases over the epoch runs and the accuracy increases.

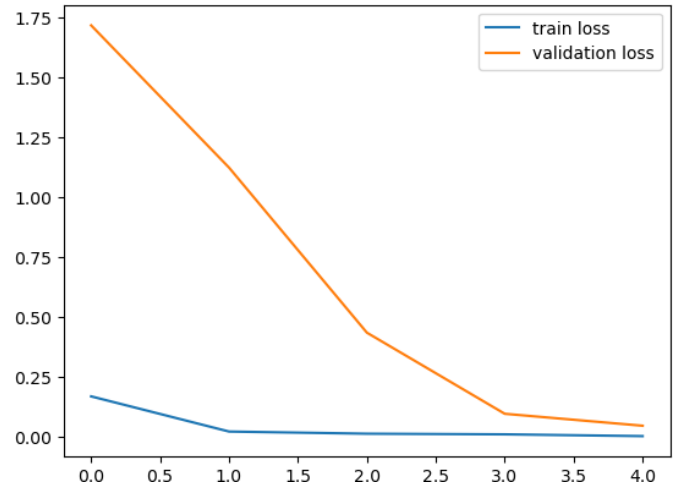


Fig. 17. Train Plot - MobileNetV2.

The validation loss tends to approach zero for the given dataset and the model has learned the training data well and the generalizing of the validation data is effective for the designed model.

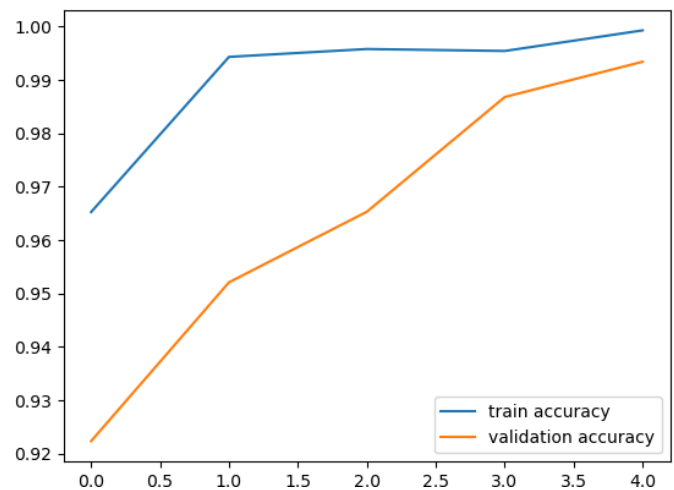


Fig. 18. Validation Loss Plot - MobileNetV2.

After the validation of the given model is conducted, the unknown data is passed to the model to check on how accurately the model classifies the images wearing a face mask or not.

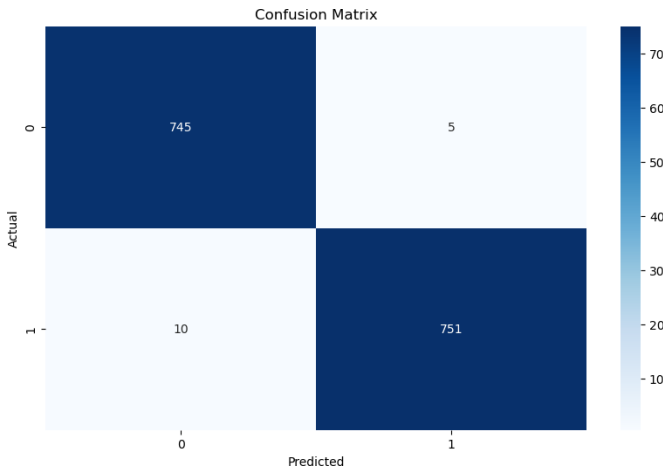


Fig. 19. Confusion Matrix - MobileNetV2.

The model has classified the images wearing face mask correctly in 751 instances and without wearing face mask in 745 instances. It has falsely classified showing False Negative as 10 and false positive in 5 cases. From the classification report generated for the model passing the test dataset, the model has classified the data 99% accuracy and precision and having higher success rates in classifying the images wearing a face mask and not wearing a face mask.

	precision	recall	f1-score	support
0	0.99	0.99	0.99	750
1	0.99	0.99	0.99	761
accuracy			0.99	1511
macro avg	0.99	0.99	0.99	1511
weighted avg	0.99	0.99	0.99	1511

Fig. 20. Classification Report - MobileNetV2.

The overall model generated has the accuracy in classifying the images at 99%.

This model has captured the features of the images accurately during the training, where it classifies the image when the face is covered with certain objects other than mask and classified correctly during testing. This is shown in Fig. 21. This model has also classified correctly between the images with and without face mask.

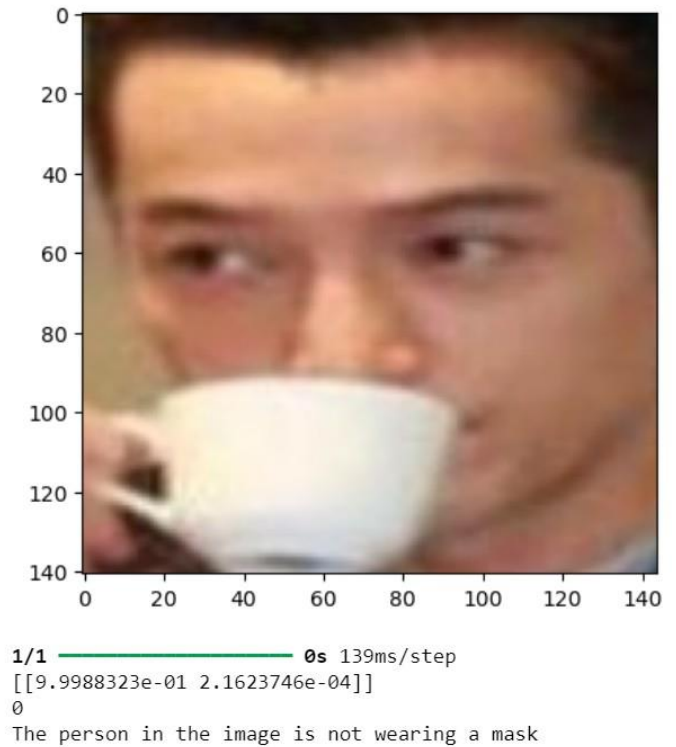


Fig. 21. Result - MobileNetV2.

V. CONCLUSION AND FUTURE WORK

For the proposed models built using Convolutional Neural Network and other 2 models employing transfer learning for learning key distinct features in image for classification, ResnetV2 and MobilenetV2. The usage of feature extraction models compared to the traditional CNN models enable the accurate training of the model with larger corpus of data using the depth of 152 layers in ResNet and 53 layers for the MobileNet model. In overall, the MobileNet model provided statistically significant results in classifying the images compared to other models built. The accuracy achieved using this model is close to 99% and the loss generated after 5 epoch iterations approaches ot zero conveying the model is a generalized model for the fully trained dataset.

Future work related to this project involves hyper parameter tuning, leading to unfreezing of some layers in the pre-trained MobileNetV2 model. Usage of ensemble models by combining multiple models including different CNN architecture and variety of MobileNetV2 model to increase the accuracy further can be tried. Employing other transfer learning methodologies like VGG, Inception and EfficientNet can provide better or faster training of the models to deploy classifying application to real-time use.

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